Pretraining Science Knowledge Base

A source-pinned knowledge base for designing data mixtures, architectures, optimizers, scaling experiments and evaluation. The original literature library and long-form manuals remain; structured source cards and canonical synthesis pages add retrieval and evidence contracts.

Read: schema and retrieval contract · PDF reading index · complete literature library · current coverage and limits.

Start from a decision

Decision	Canonical explanation	Concrete next artifact
How much model/data can our budget support?	Compute and parameter accounting	Resolved tensor/FLOP ledger and measured useful throughput
How do we build and allocate training data?	Mixture experiment design	Retained-count, phase-exposure and sampler manifests
How do width, depth, batch and LR transfer?	Hyperparameter protocol	Group-wise optimizer contract and controlled grid
What grid supports a scaling forecast?	Identifiable fits and larger holdouts	Fit-ready observations, diagnostics and untouched holdout
How do we evaluate checkpoints/stages?	Evaluation contract	Frozen task/runtime config and per-item/numerator records
What should we inspect when a result fails?	Failure diagnosis	Controlled hypothesis, intervention and before/after receipt
What exactly can we reproduce from Marin?	Marin case study	Explicit reproduction lane, input inventory and acceptance gates

The question index distinguishes documented answers from conditional plans and unresolved inputs. The hero architecture, accessible data supply, measured H100 throughput and cyber transfer remain experimental decisions.

Inspect the evidence

Follow **wiki page → claim ID → source card → original paper/code/data/result**. A source card indexes the retained original; it does not replace it or certify a paper-wide review. Confidence and reproducibility are separate fields.

- Knowledge map and prerequisites.
- Evidence standards.
- Scoped claim ledger.
- Browse topics, stages, architectures, sources, evidence kinds, and reproducibility.

Run the small checks before proposing large training

```
python scripts/kb.py query "muP batch" --type technique
python scripts/kb.py get technique-hyperparameter-transfer --follow-sources
python scripts/kb.py claim claim-marin-transfer-d1536
python scripts/kb_workbench.py --all --check
python scripts/kb.py validate --verify-artifacts --check-generated
```

The offline workbench contains seven CPU recipes for pinned public records and analytical fixtures. They are not GPU training reproductions. No claim that downloading a result reproduces its model is permitted by the schema.

Keep the literature and the detailed manuals

The saved Jiaxuan blog, supplied notes, annotated notes, Step Law, Farseer, Power Lines, and OLMo/Marin manuals remain available. The original guide portal preserves the full reading and experiment sequence.

Downloaded PDFs, extracted text, LaTeX, figures, code archives and snapshots stay under `references/`. The preservation baseline protects the existing collection. New evidence should be appended with its version and scope, not overwrite history.